Naive Lie Theory 3.4 Exercises

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November 2025

Notes

1. First let's define what is a continuous function is a topological sense: a continuous function is a function such that the preimage of an open set is open. (the being open here is not necessarily the "openness" in the sense of metric space, but here everything has a metric)

2. A fundamental theorem from topology is that the image of a path-connected space of a continuous function. Say X is path-connected and $f : X \to Y$ is continuous. Let $f(x_1) = y_1$ and $f(x_2) = y_2$. As X is path-connected, so there exists a continuous path $\gamma : I \to X$ such that $\gamma(0) = x_1$ and $\gamma(1) = x_2$. Define a path $f \circ \gamma$ in $f(X) \subseteq Y$ from y_1 to y_2. It's continuous as its components are both continuous, thus $f(X)$ is path-connected.

3.4.1

Direct calculation. We equivalently show $JB = \overline{B}J$

$$JB = \begin{pmatrix} \overline{\beta} & \overline{\alpha} \\ -\alpha & \beta \end{pmatrix} = \overline{B}J$$

3.4.2

Same calculation. Equating $JB = \overline{B}J$ gives:

$$\begin{pmatrix} e & f \\ -c & -d \end{pmatrix} = \begin{pmatrix} -\overline{d} & \overline{c} \\ -\overline{f} & \overline{e} \end{pmatrix}$$

It follows that $\overline{c} = f$ and $\overline{d} = -e$.

3.4.3

First, we compute $J_{2n}B_{2n}$. Since J_{2n} is diagonal in a block sense, it's zero unless $i = k$.

$$(J_{2n}B_{2n})_{ij} = \Sigma_{k=1}^{n}(J_{2n})_{ik}(B_{2n})_{kj} = JB_{ij}$$

J_{2n}^{-1} is also diagonal in a block sense, so

$$(J_{2n}B_{2n}J_{2n}^{-1})_{ij} = JB_{ij}J^{-1}$$

so the equation $J_{2n}B_{2n}J_{2n}^{-1} = \overline{B_{2n}}$ is equivalent to $JB_{ij}J^{-1} = \overline{B_{ij}}$

Assume B_{2n} is of the form $C(A)$, which means for every block $B_{ij} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}$. By 3.4.1 we have $JB_{ij}J^{-1} = \overline{B_{ij}}$, which is equivalent to $J_{2n}B_{2n}J_{2n}^{-1} = \overline{B_{2n}}$ as we just mentioned. Everything here is bidirectional.

3.4.4

By 3.4.3, we have

$$\det(J_{2n}B_{2n}J_{2n}^{-1}) = \det(\overline{B_{2n}})$$

LHS simplifies to:

$$\det(J_{2n})\det(B_{2n})\frac{1}{\det(J_{2n})} = \det(B_{2n})$$

RHS simplifies to $\overline{\det(B_{2n})}$. We have something equals something's conjugate, so the thing must be real.

3.4.5

By 3.4.4 we know $\det(B_{2n})$ is real. By the definition of being unitary, we have $\overline{B_{2n}}^T B_{2n} = I$. Given $\det(\overline{B_{2n}}^T) = \overline{\det(B_{2n})}$, we have $\overline{\det(B_{2n})}\det(B_{2n}) = (\det(B_{2n}))^2 = 1$, so $\det(B_{2n}) = \pm 1$.

3.4.6

Let $\det : Sp(n) \to \mathbb{C}$ be the determinant map. It's continuous because it's a polynomial. We are given that $Sp(n)$ is path-connected. Using the theorem we proved in the notes, the image $\det(Sp(n))$ is also path-connected. However, in 3.4.5 we have $\det(Sp(n)) \subseteq \{-1, 1\}$, and $\{-1, 1\}$ is obviously not path-connected. So, $\det(Sp(n))$ must be $\{-1\}$ or $\{1\}$. The easiest check is that $I \in Sp(n)$ and $\det(I) = 1$, so $\det(Sp(n)) = \{1\}$.